

ODOO 19 ENTERPRISE

Professional Multi-Project Management Guide

Covers:

- ① Multi-Project Workspace Architecture (No New Odoo Install Needed)
 - ② Database-per-Project Strategy with Multi-tenancy
 - ③ Git Organization for Multiple Projects
 - ④ Custom Module: l10n_gh_tax — Ghana Tax Localization (GRA Compliant)
 - ⑤ Ghana Tax (VAT Act 2025, Act 1151) — Full Odoo 19 Walkthrough
 - ⑥ Applying l10n_gh_tax to OmniChannel Project
 - ⑦ Changes from USD/US → GHS/Ghana for OmniChannel
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Version: Odoo 19 Enterprise Only | Country: Ghana | Currency: GHS

PART 1 — The Right Architecture: Multiple Projects on One Odoo Install

Your Core Question Answered

You asked: Do I need a whole new Odoo 19 installation for each project? NO. One Odoo 19 installation can serve unlimited projects — through multiple databases. This is the professional, industry-standard approach used by Odoo implementation partners worldwide.

1.1 Understanding How Odoo Handles Multiple Projects

Odoo's architecture separates the application server (the engine) from databases (the data). Your one Odoo 19 installation already running on port 9029 is the engine. Each project gets its own database, completely isolated from all others.

Approach	What It Means
One Odoo Install	Your <code>/opt/odoo19/odoo/</code> stays as-is. Never reinstall.
Multiple Databases	Each project = 1 database. PolicyCon has DB 'policycon'. OmniChannel gets DB 'omnichannel'. Each is 100% isolated.
Shared Custom Addons Folder	<code>/opt/odoo19/odoo-custom-addons/</code> holds ALL project modules. Odoo loads from this folder for any database.
Module Activation Per DB	Installing a module in DB 'policycon' does NOT install it in DB 'omnichannel'. Each DB controls its own installed modules.
Port Stays The Same	Still port 9029. You switch between projects by switching databases at login.

1.2 How to Create a New Database for OmniChannel

Critical Security Step

Before creating databases via the web, set a master_password in your `/etc/odoo19.conf`. Without this, anyone can create/delete databases. This is the admin password for the database manager — different from any user password.

Step 1: Set Master Password in config

```
sudo nano /etc/odoo19.conf
# Add or update this line:
admin_passwd = YourSecureMasterPasswordHere
# Save and restart:
sudo systemctl restart odoo19
```

Step 2: Access Database Manager

Go to: <http://localhost:9029/web/database/manager>
(Or from the login screen → click 'Manage Databases')

Step 3: Create New Database

Field	Value for OmniChannel
Master Password	The master password you set above
Database Name	omnichannel (use lowercase, no spaces)
Email	admin@omnidecor.com
Password	Strong admin password for this DB
Phone	Your company phone
Language	English
Country	Ghana
Demo Data	NO — never use demo data for real projects

Step 4: Switch Between Projects

When you open <http://localhost:9029> and you're not logged in, you see the database selector. Choose:

- 'policycon' → logs you into PolicyCon project
- 'omnichannel' → logs you into OmniChannel project
- Each database has its own users, modules, data — completely separate

1.3 Your Custom Addons Folder Strategy

One Folder, All Projects — That's Fine!

Your `/opt/odoo19/odoo-custom-addons/` will have ALL modules from ALL projects in it. This is perfectly normal and professional. Odoo loads the folder, but only installs modules you explicitly activate per database. Having 50 module folders in one addons path does not slow down Odoo or cause conflicts.

What your addons folder will look like:

```
# /opt/odoo19/odoo-custom-addons/
#
# PolicyCon project modules:
policycon_analyst/      ← PolicyCon
policycon_base/         ← PolicyCon
policycon_earnings/     ← PolicyCon
policycon_newsletter/   ← PolicyCon
policycon_policy_review/ ← PolicyCon
policycon_refund/       ← PolicyCon
policycon_repository/   ← PolicyCon
policycon_subscriber/   ← PolicyCon
policycon_subscription_custom/ ← PolicyCon
#
# OmniChannel project modules (from Part 12-14 of main guide):
production_efficiency_tracker/ ← OmniChannel
commission_plan/          ← OmniChannel
manufacturing_profitability/ ← OmniChannel
#
# SHARED modules (can be used by ANY project):
l10n_gh_tax/             ← Ghana Tax — works for BOTH projects!
#
```

```
# Future Project 3:  
project3_module1/  
project3_module2/
```

This is clean, professional, and scalable. When you have 7 projects with 8 modules each = 56 module folders. This is still perfectly manageable. Odoo partners regularly work with 100+ module folders in a single addons path.

1.4 Accessing Different Projects on Odoo.sh

On Odoo.sh, each project is a separate Odoo.sh project with its own repo connection and database. Your PolicyCon work is on one Odoo.sh project. OmniChannel is a new Odoo.sh project.

Odoo.sh Project	Configuration
PolicyCon	Repo: github.com/yourorg/policycon-odoo DB: auto-managed by Odoo.sh
OmniChannel (new)	Repo: github.com/yourorg/omnidecor-odoo DB: auto-managed by Odoo.sh

Each Odoo.sh project is fully independent — separate staging, separate production, separate URLs.

PART 2 — Git Organization for Multiple Projects (The Professional Way)

2.1 The Problem You're Sensing

You asked: 'If I have 7 projects each with 8 modules, won't git become unmanageable from one folder?' — Correct instinct. The answer is: ONE GIT REPO PER PROJECT, not one repo for all projects.

2.2 The Golden Rule: One Repo Per Project

Approach	Why
ONE repo per project (CORRECT)	PolicyCon changes don't pollute OmniChannel git history. Each project can have its own CI/CD pipeline on Odoo.sh. Different teams can have different access. Clean commit history per project.
One repo for all projects (WRONG)	Merging staging → production for PolicyCon would also deploy OmniChannel code. Odoo.sh doesn't support this pattern. Commit history becomes a nightmare. Access control is impossible per project.

2.3 Your Repository Structure

```
# GitHub Organization: github.com/YourOrg/
#
# PolicyCon project repo:
github.com/YourOrg/policycon-odoo/
├─ policycon_analyst/
├─ policycon_base/
├─ policycon_earnings/
└─ ... (all 9 policycon modules)
#
# OmniChannel project repo:
github.com/YourOrg/omnidector-odoo/
├─ production_efficiency_tracker/
├─ commission_plan/
├─ manufacturing_profitability/
└─ ... (OmniChannel custom modules)
#
# Ghana Tax Localization (SHARED LIBRARY repo):
github.com/YourOrg/odoo-gh-tax/
└─ ll0n_gh_tax/      ← Ghana tax module, reusable across all projects
#
# Future projects get their own repos:
github.com/YourOrg/project3-odoo/
github.com/YourOrg/project4-odoo/
```

2.4 How to Use the Shared Ghana Tax Module Across Projects

Git Submodules — The Professional Solution

For shared modules like `l10n_gh_tax` that multiple projects use, we use Git Submodules. Each project repo includes the shared module as a submodule — it pulls from the shared repo but stays versioned. If you update `l10n_gh_tax`, each project can pull the update independently.

Option A: Git Submodule (Recommended for shared modules)

```
# In the OmniChannel repo, add l10n_gh_tax as a submodule:
cd /opt/odoo19/odoo-custom-addons
git checkout staging
git submodule add https://github.com/YourOrg/odoo-gh-tax.git l10n_gh_tax
git commit -m 'feat: add Ghana tax localization as submodule'
git push origin staging
#
# In PolicyCon repo, do the same:
cd /path/to/policycon-odoo
git submodule add https://github.com/YourOrg/odoo-gh-tax.git l10n_gh_tax
git commit -m 'feat: add Ghana tax localization as submodule'
```

Option B: Simple Copy (Simpler but doesn't auto-sync updates)

```
# Just copy the module into each project's addons folder.
# When l10n_gh_tax updates, manually copy updates to each project.
cp -r /path/to/l10n_gh_tax /path/to/project-addons/
# Simpler but requires manual sync when the shared module updates.
```

2.5 Local Development: Managing Multiple Project Repos

Your local setup:

```
# Your main addons folder has SYMLINKS or all modules together:
ls /opt/odoo19/odoo-custom-addons/
#
# Structure approach 1: All projects share one addons folder (SIMPLE):
# Just clone each project's modules directly into the addons folder.
# Git will show it as ONE big repo – this works fine locally.
#
# Structure approach 2: Separate folders with symlinks (CLEANER):
/opt/odoo19/
├── odoo-custom-addons/          ← Odoo reads THIS folder
│   ├── l10n_gh_tax -> ../projects/odoo-gh-tax/l10n_gh_tax
│   ├── policycon_base -> ../projects/policycon-odoo/policycon_base
│   └── production_efficiency_tracker -> ../projects/omnidecor-odoo/prod_eff..
├── projects/
│   ├── policycon-odoo/          ← PolicyCon git repo
│   ├── omnidecor-odoo/          ← OmniChannel git repo
│   └── odoo-gh-tax/              ← Ghana Tax shared repo
```

Creating symlinks (Structure approach 2):

```
sudo su - odoo19
mkdir -p /opt/odoo19/projects
#
# Clone OmniChannel repo
cd /opt/odoo19/projects
```

```

git clone https://github.com/YourOrg/omnidecor-odoo.git
git clone https://github.com/YourOrg/odoo-gh-tax.git
#
# Create symlinks in the addons folder
cd /opt/odoo19/odoo-custom-addons
ln -sf /opt/odoo19/projects/omnidecor-odoo/production_efficiency_tracker .
ln -sf /opt/odoo19/projects/omnidecor-odoo/commission_plan .
ln -sf /opt/odoo19/projects/odoo-gh-tax/l10n_gh_tax .
#
# Now Odoo sees all modules in odoo-custom-addons/
# but each is managed by its own git repo

```

2.6 Odoo.sh Config File — Multiple Addons Paths

Odoo.sh automatically manages addons paths. But on your local server, update `/etc/odoo19.conf` if you use the separate folders approach:

```

sudo nano /etc/odoo19.conf
#
[options]
admin_passwd = YourMasterPassword
db_host = False
db_port = False
db_user = odoo19
db_password = False
# addons_path lists ALL folders Odoo should look in:
addons_path = /opt/odoo19/odoo/addons,/opt/odoo19/odoo-custom-addons
# If you use the separate projects folder structure:
# addons_path = /opt/odoo19/odoo/addons,/opt/odoo19/odoo-custom-
# addons,/opt/odoo19/projects/odoo-gh-tax
logfile=/var/log/odoo19/odoo19.log
xmlrpc_port = 9029

```

After any addons_path change:

```

sudo systemctl restart odoo19
# Then in each database where you want to use new modules:
# Settings → Technical → Apps → Update Apps List

```

PART 3 — Ghana Taxes in Odoo 19: Complete Understanding

Legal Reference

All Ghana tax rates and rules in this guide are based on the Value Added Tax Act, 2025 (Act 1151) effective 1 January 2026, and the Income Tax Act, 2015 (Act 896). COVID-19 Health Recovery Levy has been ABOLISHED from January 1, 2026.

3.1 Ghana Tax Landscape — What You're Dealing With

Tax Type	Details (from GRA, Act 1151, effective Jan 2026)
VAT — 15%	Standard rate on taxable supplies of goods and services in Ghana. Same base as NHIL and GETFund (recoupled from Jan 2026).
NHIL — 2.5%	National Health Insurance Levy. Now claimable as input tax (recoupled). Same base as VAT.
GETFund — 2.5%	Ghana Education Trust Fund Levy. Now claimable as input tax (recoupled). Same base as VAT.
Effective Total — 20%	VAT 15% + NHIL 2.5% + GETFund 2.5% = 20% on the same taxable value.
COVID-19 Levy — ABOLISHED	Removed from 1 January 2026. Do NOT create this tax in Odoo 19.
WHT — Services 7.5%	Withheld from payment to service vendors. Remit to GRA by 15th of next month.
WHT — Goods 3%	Withheld from payment to goods suppliers (above GHS 2,000 threshold).
WHT — Works 5%	Withheld from payments for construction works, contracts.
WHT — Management Fees 20%	Director fees, management fees paid to resident individuals.
WHT — Rent 8%/15%	8% residential, 15% commercial. Withheld from rent payments.
WHT — Investment Returns 8%	Dividends and interest — 8% withheld.
VAT WHT Agent — 7%	GRA-appointed agents (government, banks) withhold 7% of taxable value when paying you.

3.2 How the 20% Works (Critical Calculation)

IMPORTANT: Old System vs New System (2026)

OLD (before 2026): NHIL and GETFund were added to cost FIRST, then VAT was calculated on the inflated amount (cascading = more expensive). NEW (from 2026, Act 1151): ALL THREE taxes (VAT 15% + NHIL 2.5% + GETFund 2.5%) are calculated on the SAME base. This is how our l10n_gh_tax module is configured — all three as parallel percentage taxes in a group.

Example: Invoice for GHS 1,000 service

Tax Component	Calculation
Taxable Amount (base)	GHS 1,000.00
VAT 15%	$GHS\ 1,000 \times 15\% = GHS\ 150.00 \rightarrow$ posted to 2310 VAT Output
NHIL 2.5%	$GHS\ 1,000 \times 2.5\% = GHS\ 25.00 \rightarrow$ posted to 2311 NHIL Output
GETFund 2.5%	$GHS\ 1,000 \times 2.5\% = GHS\ 25.00 \rightarrow$ posted to 2312 GETFund Output
Total Tax	GHS 200.00
Invoice Total	GHS 1,200.00
Journal Entry (Dr)	Dr Accounts Receivable 1,200
Journal Entry (Cr)	Cr Revenue 1,000 Cr VAT Output 150 Cr NHIL Output 25 Cr GETFund Output 25

3.3 Withholding Tax — How It Actually Works in Odoo 19

WHT is NOT charged on the invoice — it is DEDUCTED at payment time

This is the most misunderstood part of Ghana WHT in Odoo. You enter the vendor bill for the FULL amount. Then when you pay, you pay LESS (net of WHT). The difference goes to your WHT Payable account. You later pay that to GRA.

Scenario: IT Consulting Service Invoice — GHS 10,000

1. Vendor (XYZ Tech) submits invoice for GHS 10,000 + 20% VAT = GHS 12,000 total
2. You enter the vendor bill in Odoo: GHS 10,000 + VAT+Levies 20% = GHS 12,000
3. Post the bill
4. Click 'Register Payment'
5. You type: Amount = GHS 10,750 (which is GHS 12,000 minus WHT 7.5% of GHS 10,000)
6. In the payment difference, select 'Mark as fully paid' with counterpart account: WHT Payable — Services (account 2320)
7. Result: Bill settled. Vendor got GHS 10,750. You owe GRA GHS 750 in WHT Payable.
8. You also owe GRA: Input VAT GHS 1,500 + Input NHIL GHS 250 + Input GETFund GHS 250 (claimable)
9. Generate WHT Certificate for XYZ Tech (Ghana Tax → WHT Certificates → New)
10. By 15th of next month: pay GRA GHS 750 WHT → clear account 2320

3.4 VAT Return Process in Odoo 19 (Monthly)

11. Confirm ALL invoices and bills for the month are POSTED (no drafts)
12. Reconcile all bank payments
13. Go to: Accounting → Reporting → Tax Return
14. Select period: e.g., February 2026
15. Review: Output taxes (what you collected from customers) vs Input taxes (what you paid to vendors)
16. Click 'Closing Entry' → this creates a journal entry that net-settles your tax accounts
17. Post the closing entry

18. Calculate net payable: Total Output Tax - Total Input Tax = Amount to Pay GRA
19. File on GRA's online portal using Odoo's report as source of truth
20. Create payment: Accounting → Vendors → Payments → Pay 'Ghana Revenue Authority (GRA)' vendor
21. Counterpart account: 2330 GRA Tax Payable — Net Settlement
22. Mark as submitted in Odoo

VAT Filing Deadline

VAT return (VAT + NHIL + GETFund) is due by the LAST WORKING DAY of the following month.
Example: January 2026 returns → due last working day of February 2026. WHT returns → 15th of the following month.

PART 4 — l10n_gh_tax Module: Complete Explanation

4.1 Module Overview

The module's technical name is l10n_gh_tax (following Odoo's naming convention: l10n = localization, gh = Ghana). When installed in any Odoo 19 database, it automatically creates everything needed for Ghana GRA compliance.

4.2 Module File Structure

```
l10n_gh_tax/
├── __init__.py           ← Imports models and hooks
├── __manifest__.py       ← Module metadata, dependencies, data files list
├── hooks.py             ← post_init_hook: runs once on install
├── README.md            ← Documentation
├──
├── models/
│   ├── __init__.py      ← Imports all model files
│   ├── account_move.py  ← Extends invoices: TIN, VAT invoice no., WHT fields
│   ├── account_tax.py   ← Extends taxes: gh_tax_type, gh_is_withholding_tax fields
│   ├── res_partner.py   ← Extends contacts: Ghana TIN, VAT registered, WHT agent
│   └── wht_certificate.py ← New model: Ghana WHT Certificate (gh.wht.certificate)
├── data/                ← ALL data loaded automatically on install
│   ├── account_chart_template_data.xml ← Activates GHS currency
│   ├── account_tax_group_data.xml      ← 6 tax groups (VAT, NHIL, GETFund, WHT, VAT WHT,
Exempt)
│   ├── account_account_data.xml        ← 13 Ghana tax accounts (2310-2330, 1310-1320)
│   ├── account_tax_data.xml            ← 16 taxes + WHT certificate sequence
│   ├── account_fiscal_position_data.xml ← 5 fiscal positions
│   └── res_partner_gra_data.xml         ← GRA as a vendor
├── views/
│   ├── account_move_views.xml          ← Ghana Tax tab on invoices/bills + WHT Certificate views
│   ├── res_partner_views.xml           ← Ghana Tax tab on customer/vendor forms
│   └── menu_items.xml                  ← 'Ghana Tax (GRA)' menu under Accounting
├── security/
│   └── ir.model.access.csv             ← Access rights for WHT Certificate model
├── report/
│   ├── withholding_certificate_report.xml ← PDF WHT Certificate template
│   └── gra_tax_summary_report.xml         ← Placeholder for GRA summary
```

4.3 What Gets Created on Install — Full List

4.3.1 Tax Accounts Created

Account	Code & Type
VAT Output Payable — GRA (15%)	2310 Liability — you owe this to GRA from sales

VAT Input Receivable — GRA (15%)	1310 Asset — GRA owes you this from purchases
NHIL Output Payable — GRA (2.5%)	2311 Liability
NHIL Input Receivable — GRA (2.5%)	1311 Asset — now claimable from Jan 2026
GETFund Output Payable — GRA (2.5%)	2312 Liability
GETFund Input Receivable — GRA (2.5%)	1312 Asset — now claimable from Jan 2026
WHT Payable — Services (7.5%)	2320 Liability Reconcilable
WHT Payable — Goods (3%)	2321 Liability Reconcilable
WHT Payable — Works/Contracts (5%)	2322 Liability Reconcilable
WHT Payable — Management Fees (20%)	2323 Liability Reconcilable
WHT Payable — Rent (8/15%)	2324 Liability Reconcilable
WHT Payable — Investment Returns	2325 Liability Reconcilable (Dividends/Interest/Royalties)
WHT Receivable — Tax Credit	1320 Asset Reconcilable — your WHT credits
VAT WHT Payable — Agent 7%	2313 Liability
GRA Tax Payable — Net Settlement	2330 Liability Reconcilable — for final VAT payment to GRA

4.3.2 Taxes Created

Tax Name	Type	Rate & Account
VAT 15% (Sales)	sale	15% → 2310 VAT Output Payable
NHIL 2.5% (Sales)	sale	2.5% → 2311 NHIL Output Payable
GETFund 2.5% (Sales)	sale	2.5% → 2312 GETFund Output Payable
VAT+Levies 20% Standard (Sales)	sale	GROUP: VAT15+NHIL2.5+GETFund2.5 ← USE THIS on products
VAT 15% (Purchase)	purchase	15% → 1310 VAT Input Receivable
NHIL 2.5% (Purchase)	purchase	2.5% → 1311 NHIL Input Receivable
GETFund 2.5% (Purchase)	purchase	2.5% → 1312 GETFund Input Receivable
VAT+Levies 20% Standard (Purchase)	purchase	GROUP: all 3 input taxes ← USE THIS on products
VAT 0% Zero-Rated	sale	0% — for exports, textiles, sanitary towels
VAT Exempt (Sales)	sale	0% — for exempt goods/services
VAT Exempt (Purchase)	purchase	0% — for non-VAT registered vendors

WHT Services 7.5%	purchase	-7.5% → 2320 (negative = reduces your payable to vendor)
WHT Goods 3%	purchase	-3% → 2321
WHT Works 5%	purchase	-5% → 2322
WHT Management Fees 20%	purchase	-20% → 2323
WHT Rent Residential 8%	purchase	-8% → 2324
WHT Rent Commercial 15%	purchase	-15% → 2324
WHT Dividends 8%	purchase	-8% → 2325
WHT Interest 8%	purchase	-8% → 2325
WHT Royalties 15%	purchase	-15% → 2325
WHT Non-Resident 25%	purchase	-25% → 2320
VAT WHT Agent 7% (Sales)	sale	-7% → 2313 (shows on your sales invoice when selling to WHT agents)

4.3.3 Fiscal Positions Created

Fiscal Position	When to Use
Ghana — Standard VAT Registered	DEFAULT. For all VAT-registered customers and vendors in Ghana. All standard taxes apply.
Ghana — Zero-Rated / Exports	For export customers, textiles, sanitary towels. Maps 20% group → 0% zero-rated.
Ghana — VAT Exempt	For exempt goods/services (food, medical, education). Maps 20% → exempt 0%.
Ghana — Non-VAT Registered Supplier	For vendors NOT registered for VAT (threshold GHS 750,000 for goods). No input VAT charged.
Ghana — VAT Withholding Agent Customer	For GRA-appointed WHT agents (government, banks). They withhold 7% of your VAT.

4.4 Fields Added to Existing Models

On Contacts (res.partner):

Field	Purpose
Ghana TIN	GRA Taxpayer ID. Format: C0012345678 (company) or P0012345678 (person). Validated by regex.
VAT Registered (GRA)	True/False. If True, TIN becomes required on posted invoices.
GRA VAT Withholding Agent	If True, this customer is a GRA-appointed WHT agent — they'll withhold 7% of your VAT.
WHT Agent Certificate No.	The GRA certificate number for the withholding agent.
SSNIT Number	Social Security number (for employee records).

Tax Exemption Reference	GRA exemption certificate number.
Default WHT Rate	Suggested WHT type when paying this vendor (Services 7.5%, Goods 3%, etc.).

On Invoices & Bills (account.move):

Field	Purpose
TIN (Ghana)	Auto-filled from customer/vendor TIN. Shows on invoice.
GRA VAT Invoice No.	Sequential VAT invoice number as registered with GRA. Required on standard-rated invoices.
WHT Certificate No.	Certificate number from vendor WHT certificate (on vendor bills).
WHT Amount Deducted	Computed: total WHT on this bill.
VAT Withholding Agent	Inherited from customer — shows if they're a WHT agent.
VAT Withheld (7%)	Computed: 7% of taxable value when customer is a WHT agent.

4.5 The WHT Certificate Model

The `gh.wht.certificate` model is the most important unique feature of this module. When you withhold tax from a vendor, Ghanaian law requires you to issue them a WHT Certificate as proof. This model tracks every certificate you issue.

Feature	How It Works
Auto-numbered certificates	Format: WHT/2026/0001 (sequence auto-increments)
Links to vendor bill & payment	Full audit trail: certificate → bill → payment
Computes rates automatically	Select WHT type (Services/Goods/Works) → rate and amount auto-calculate
State workflow	Draft → Issued (to vendor) → Remitted (to GRA)
PDF certificate print	One-click PDF generation with full GRA-compliant format including your company info, vendor info, amounts, WHT details, and signature fields
Monthly summary	<code>get_wht_summary_for_period()</code> method gives you aggregated WHT by type for the monthly return
Validation	Cannot issue without vendor TIN. Prompts to add TIN first.

4.6 The `post_init_hook`

When the module is first installed, the `post_init_hook` in `hooks.py` automatically:

- Sets company currency to GHS (Ghana Cedi)
- Sets fiscal country to Ghana

- Sets tax rounding method to 'round per line' (GRA standard)
- Logs a reminder to set tax return periodicity to Monthly

This means after installing `l10n_gh_tax`, your database is immediately ready for Ghana tax operations — no manual configuration needed.

PART 5 — Installing l10n_gh_tax: Step-by-Step

5.1 Adding Module to Your Server

23. Copy the l10n_gh_tax folder into your custom addons:

```
sudo su - odoo19
cd /opt/odoo19/odoo-custom-addons/
# If using zip:
# unzip l10n_gh_tax_module.zip
# Module folder should now be: /opt/odoo19/odoo-custom-addons/l10n_gh_tax/
```

24. Restart Odoo service:

```
sudo systemctl restart odoo19
```

25. Update Apps list in the target database:

Go to: Settings → Apps → Update Apps List (top right button) — you need Developer Mode on for this button to be visible.

To enable Developer Mode: Settings → General Settings → scroll to Developer Tools → Activate Developer Mode

26. Install the module:

Apps → remove 'Apps' filter → search 'Ghana Tax' → click Install on 'Ghana Tax Localization (GRA Compliant)'

5.2 Or Install via CLI (Faster for Testing)

```
sudo systemctl stop odoo19
sudo -u odoo19 /opt/odoo19/odoo-venv/bin/python3 \
  /opt/odoo19/odoo/odoo-bin \
  -c /etc/odoo19.conf \
  -d omnichannel \
  -i l10n_gh_tax \
  --stop-after-init
sudo systemctl start odoo19
```

5.3 Post-Install Verification Checklist

Check	Where to Verify
GHS currency is active	Accounting → Configuration → Currencies → GHS should show as active
13+ accounts created	Accounting → Configuration → Chart of Accounts → search '2310' or '1310'
Taxes created	Accounting → Configuration → Taxes → filter by 'Ghana' — should see 20+ taxes
Fiscal Positions created	Accounting → Configuration → Fiscal Positions → should see 5 Ghana positions
GRA vendor exists	Accounting → Vendors → search 'Ghana Revenue Authority'

Ghana Tax tab on invoices	Create a draft invoice → look for 'Ghana Tax (GRA)' tab
Ghana Tax tab on contacts	Open any contact → look for 'Ghana Tax (GRA)' tab
WHT Certificate menu	Accounting → Ghana Tax (GRA) → WHT Certificates

5.4 Setting Up Products for Ghana

27. Go to each product → Sales tab → Customer Taxes:

Remove any old tax → Add: 'VAT+Levies 20% Standard (Sales) — GRA Act 1151'

28. Go to each product → Purchase tab → Vendor Taxes:

Remove any old tax → Add: 'VAT+Levies 20% Standard (Purchase) — GRA Act 1151'

29. Set on product categories for bulk assignment:

Inventory → Configuration → Product Categories → each category → Accounting Properties → set default taxes

5.5 Setting Up Customers for Ghana

30. Open customer record → Ghana Tax (GRA) tab

31. Enter Ghana TIN (format: C0012345678 for companies)

32. Tick 'VAT Registered (GRA)' if they are VAT-registered

33. Tick 'GRA VAT Withholding Agent' if they are a government agency or GRA-appointed WHT agent

34. Go to Sales & Purchase tab → set Fiscal Position = 'Ghana — Standard VAT Registered'

5.6 Reusing the Module on PolicyCon

To use l10n_gh_tax on your PolicyCon database:

35. The module is already in your addons folder (shared)

36. Switch database to 'policycon' database

37. Apps → Update Apps List → search 'Ghana Tax' → Install

38. The same accounts, taxes, and fiscal positions are created in the PolicyCon database independently

No conflicts. No duplication issues. Each database gets its own independent copy of the tax setup.

PART 6 — OmniChannel Project: Changes for Ghana

6.1 Summary of Changes Needed

OmniChannel was designed with USA settings in the main project guide. Here are ALL the changes to make it Ghana-correct. The custom modules (Production Efficiency Tracker, Commission Plan, Manufacturing Profitability) don't need code changes — only Odoo configuration changes.

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6.2 Company Settings Changes

Setting (was)	Change to (Ghana)
Company: OmniDecor Furniture Ltd	Keep name OR rename to match actual Ghana-registered company name
Country: United States	→ Country: Ghana
State: Texas	→ Remove / leave blank (Ghana has Regions, not States in Odoo by default)
City: Houston	→ Your actual city: e.g., Accra / Tema / Kumasi / Takoradi
ZIP: 77001	→ Remove or use Ghana postal code (Ghana uses 10-digit digital addresses)
Phone: +1 (713)...	→ +233 ... (Ghana country code)
Currency: USD	→ GHS (Ghana Cedi) — auto-set by l10n_gh_tax hook
Fiscal Country: United States	→ Ghana — auto-set by l10n_gh_tax hook
Tax: US Sales Tax 8.25%	→ REMOVE — replace with Ghana VAT+Levies 20%
Chart of Accounts: US - Generic	→ Keep OR use manual Ghana CoA. l10n_gh_tax adds tax accounts on top.
Tax Return Periodicity	→ Monthly (GRA requirement for VAT-registered entities)

6.3 Tax Changes

Remove ALL US taxes. Replace with Ghana taxes from l10n_gh_tax.

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Taxes to REMOVE (from OmniChannel database):

- US Sales Tax - Standard (8.25%) → DELETE
- US Sales Tax - Exempt → DELETE
- Import Duty Placeholder → DELETE (use Ghana Landed Costs instead)

Taxes to SET on all products (after l10n_gh_tax install):

Product Tax Setting	Ghana Tax to Set
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Customer Taxes (Sales)	VAT+Levies 20% Standard (Sales) — GRA Act 1151
Vendor Taxes (Purchase)	VAT+Levies 20% Standard (Purchase) — GRA Act 1151

6.4 Fiscal Position Changes

OmniChannel Fiscal Position (was)	Ghana Equivalent
Domestic B2B	→ Ghana — Standard VAT Registered
Domestic B2C	→ Ghana — Standard VAT Registered (same rate as B2B now)
Export - Tax Exempt	→ Ghana — Zero-Rated / Exports
EU VAT	→ Create: International Export (no Ghana VAT) if needed

6.5 Pricing Changes

In Ghana, both B2B and B2C pay the same 20% VAT rate (unlike USA where it varies by state). So the B2B vs B2C distinction for taxes disappears. You keep pricelists for wholesale discounts, but the tax rate is uniform.

Pricelist (was)	Ghana Version
Standard B2C Retail (USD)	→ Standard Retail (GHS) Tax: VAT+Levies 20% included or excluded (your choice)
B2B Wholesale 10% off (USD)	→ Ghana Wholesale 10% off (GHS) Same tax rate 20%
B2B Premium 15% off (USD)	→ Ghana Premium 15% off (GHS)
Export USD	→ Export USD (keep USD for international) Fiscal Position: Zero-Rated
Export EUR	→ Export EUR (keep EUR for EU customers) Fiscal Position: Zero-Rated

6.6 Simulated Data — Ghana Version

6.6.1 Ghana Customers

Customer Name	Details
Accra Hotels Group Ltd	TIN: C0012345001 VAT Registered Accra, Ghana B2B Wholesale
Kumasi Interiors Ltd	TIN: C0012345002 VAT Registered Kumasi, Ghana B2B Wholesale
Ministry of Education Ghana	TIN: G0012345003 VAT WHT Agent Accra Government
Kofi Mensah (POS Customer)	TIN: None Individual retail customer
Abena Appiah	TIN: None eCommerce customer

Nigeria Furniture Import Co	Non-Ghana Export USD Zero-rated
UK Furnishings Ltd	London, UK Export GBP Zero-rated

6.6.2 Ghana Vendors

Vendor Name	Details + WHT Rate
Kumasi Timber Merchants Ltd	TIN: C0098765001 VAT Registered Oak/Mahogany lumber WHT: Goods 3%
AccraFoam Ghana Ltd	TIN: C0098765002 VAT Registered Foam, upholstery WHT: Goods 3%
GhanaTech IT Services	TIN: C0098765003 VAT Registered IT support WHT: Services 7.5%
Accra Freight & Logistics	TIN: C0098765004 VAT Registered Shipping WHT: Services 7.5%
Tema Port Services	TIN: C0098765005 VAT Registered Import handling WHT: Works 5%
Ghana Revenue Authority (GRA)	Already created by l10n_gh_tax module Tax remittances

6.7 Accounting Chart of Accounts — Ghana Note

In the original OmniChannel guide, US account codes (1100, 2100, etc.) were used. The l10n_gh_tax module adds Ghana-specific tax accounts (2310-2330, 1310-1320) on top. Your other accounts (revenue, expenses, AR, AP) can keep their codes or you can renumber. What matters is that the TAX accounts are properly set in each tax's repartition lines — which l10n_gh_tax handles automatically.

Key accounts to verify in OmniChannel DB after installing l10n_gh_tax:

Account	Purpose in OmniChannel Ghana
2310 VAT Output Payable	Collects 15% VAT from all sales invoices (furniture sales to Ghana customers)
2311 NHIL Output Payable	Collects 2.5% NHIL from sales
2312 GETFund Output Payable	Collects 2.5% GETFund from sales
1310 VAT Input Receivable	Recovers VAT paid on timber, foam, hardware purchases
1311 NHIL Input Receivable	Recovers NHIL paid on purchases (now claimable from Jan 2026)
1312 GETFund Input Receivable	Recovers GETFund paid on purchases (now claimable from Jan 2026)
2320 WHT Payable — Services	When paying GhanaTech IT Services: withhold 7.5% here
2330 GRA Net Settlement	After VAT closing entry: net VAT payable lands here for final GRA payment

6.8 Sample Ghana Invoice (OmniChannel — Dining Table Sets)

Scenario: Accra Hotels Group Ltd orders 10x OD-DT-001 Dining Table Sets at GHS 5,500 each

Invoice Line	Amount (GHS)
10x OD-DT-001 Solid Mahogany Dining Table Set @ GHS 5,500	GHS 55,000.00
Less: Wholesale 10% discount	(GHS 5,500.00)
Subtotal (Taxable Amount)	GHS 49,500.00
VAT 15%	GHS 7,425.00
NHIL 2.5%	GHS 1,237.50
GETFund 2.5%	GHS 1,237.50
TOTAL TAX (20%)	GHS 9,900.00
INVOICE TOTAL	GHS 59,400.00

Journal Entry Posted (when invoice is validated):

Dr	Accounts Receivable	GHS 59,400.00
Cr	Sales Revenue – B2B	GHS 49,500.00
Cr	VAT Output Payable (2310)	GHS 7,425.00
Cr	NHIL Output Payable (2311)	GHS 1,237.50
Cr	GETFund Output Payable (2312)	GHS 1,237.50

6.9 Monthly Ghana Tax Routine for OmniChannel

Activity	Deadline / Action
Post all sales invoices for the month	Before month-end
Post all vendor bills (timber, foam, hardware, services)	Before month-end
Process all WHT deductions (pay vendors NET, issue WHT certs)	At payment time; cert issued same day
Remit WHT to GRA (Services 7.5%, Goods 3%, Works 5%)	15th of following month
Run Tax Return in Odoo (output vs input)	Last week of following month
File VAT return on GRA portal	Last working day of following month
Pay net VAT to GRA → clear account 2330	Same day as filing or before deadline
Mark submitted in Odoo	Immediately after GRA portal filing

PART 7 — Quick Reference & Summary

7.1 The Three Questions — Answered Definitively

Q1: Do I need a new Odoo 19 installation for OmniChannel?

NO. One installation serves all projects via multiple databases. Create a new database called 'omnichannel' in the database manager at /web/database/manager.

Q2: How do I avoid PolicyCon and OmniChannel affecting each other?

They're in separate databases — they cannot affect each other. Installing a module in 'omnichannel' database does absolutely nothing to 'policycon' database. Data is completely isolated. Users are separate. Settings are separate.

Q3: One git repo for everything or separate repos?

Separate repos. One repo per project. For shared modules like l10n_gh_tax, use a dedicated shared repo and add it as a git submodule to each project. Never one mega-repo for all projects.

7.2 Ghana Tax Module — At a Glance

What you select on an invoice	What Odoo does behind the scenes
VAT+Levies 20% Standard (Sales)	Creates 3 tax lines: VAT 15% → 2310, NHIL 2.5% → 2311, GETFund 2.5% → 2312
VAT+Levies 20% Standard (Purchase)	Creates 3 input tax lines going to 1310, 1311, 1312 (all claimable from Jan 2026)
WHT Services 7.5% on vendor bill	Creates a negative tax line: reduces what you owe vendor by 7.5%, posts to 2320
Zero-Rated 0%	No tax charged. Supplier still claims input VAT on their costs.
Fiscal Position: Zero-Rated/Exports	Automatically maps 20% group → 0% on that customer's invoices

7.3 WHT Certificate Workflow — Quick Reference

39. Receive vendor invoice (e.g., GhanaTech: GHS 10,000 + 20% VAT = GHS 12,000)
40. Enter vendor bill in Odoo with VAT+Levies Purchase tax
41. Post the bill
42. Register Payment → Amount: GHS 11,250 (GHS 12,000 - 7.5% of 10,000 = GHS 750 withheld)
43. Write-off difference (GHS 750) to account 2320 WHT Payable — Services
44. Accounting → Ghana Tax → WHT Certificates → New
45. Enter: vendor, type=Services 7.5%, gross=GHS 10,000, date, description
46. Click Issue → PDF certificate generated → email/print to GhanaTech
47. By 15th of following month: pay GRA GHS 750, mark certificate as Remitted

7.4 GRA Filing Deadlines Summary

Tax / Obligation	Deadline
VAT + NHIL + GETFund return & payment	Last working day of the following month
Withholding Tax (all types) return & payment	15th of the following month
PAYE (employee income tax)	15th of the following month
Annual Corporate Income Tax return	4 months after end of accounting year
Annual PAYE reconciliation	30 April of the following year
WHT property transactions	15 days after transaction completion
Penalties for late filing	GHS 500 + GHS 10 per additional day
Late payment interest	125% of Bank of Ghana Monetary Policy Rate, compounded monthly
Document retention	6 years minimum (WHT certificates, invoices, returns)

7.5 Module Deployment Summary (Git)

What to push	Which repo / branch
l10n_gh_tax/ module folder	github.com/YourOrg/odoo-gh-tax → main branch
OmniChannel custom modules (production_efficiency_tracker, etc.)	github.com/YourOrg/omnidecor-odoo → staging first, then main
PolicyCon modules (policycon_analyst, etc.)	github.com/YourOrg/policycon-odoo → staging first, then main
l10n_gh_tax as submodule in OmniChannel	omnidecor-odoo → git submodule add odoo-gh-tax.git
l10n_gh_tax as submodule in PolicyCon	policycon-odoo → git submodule add odoo-gh-tax.git
NEVER push	Core Odoo source (/opt/odoo19/odoo/), database files, /etc/odoo19.conf, filestore/

— End of Multi-Project & Ghana Tax Guide —

l10n_gh_tax | Version 19.0.1.0.0 | GRA Act 1151 Compliant | January 2026